28/4/2026

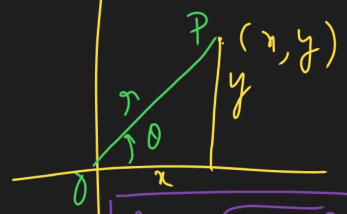
Recall

$$s = \begin{cases} \displaystyle\int_{x_1}^{x_2} \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx \\ \displaystyle\int_{y_1}^{y_2} \sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy \end{cases}$$

$$\boxed{\begin{aligned} P &= (x,y) = x+iy = z \\ &= (r,\theta) = re^{i\theta} \end{aligned}}$$

Cartesian coords

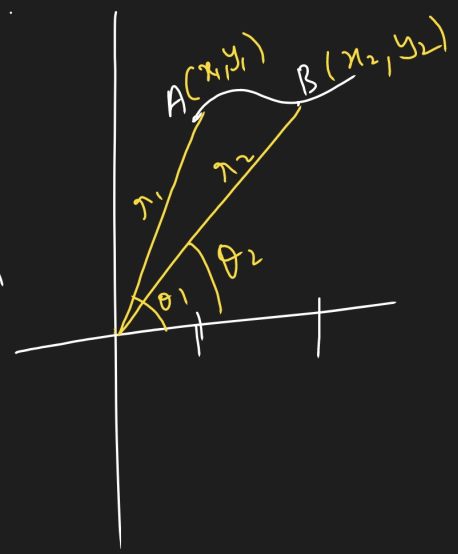
$$\boxed{\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \end{aligned}}$$



$$\boxed{\begin{aligned} r &= \sqrt{x^2+y^2} \\ \theta &= \tan^{-1} y/x \end{aligned}}$$

Length of a curve in polar coordinates (r,θ)

$$s = \begin{cases} \displaystyle\int_{\theta=\theta_1}^{\theta_2} \sqrt{r^2+\left(\frac{dr}{d\theta}\right)^2}\, d\theta \\ \displaystyle\int_{r=r_1}^{r_2} \sqrt{1+r^2\left(\frac{d\theta}{dr}\right)^2}\, dr \end{cases}$$



Exercise

Identify the foll. curves.

1. $r = a\cos\theta$

$r = \sqrt{x^2+y^2}$; $x = r\cos\theta$; $y = r\sin\theta$

$\Rightarrow r^2 = x^2+y^2$

Now $r = a\cos\theta$

$\Rightarrow r^2 = a^2\cos^2\theta$

$= a.(a\cos\theta).\cos\theta$

$\Rightarrow x^2+y^2 = a.(r)\cos\theta$

$\Rightarrow x^2+y^2 = a.r\cos\theta$

$\Rightarrow x^2+y^2 = a.x$

$\Rightarrow x^2 - ax + y^2 = 0$

$\Rightarrow \left(x-\frac{a}{2}\right)^2 - \frac{a^2}{4} + y^2 = 0$

$\Rightarrow \left(x-\frac{a}{2}\right)^2 + y^2 = \left(\frac{a}{2}\right)^2$

Which represents a circle with center $\left(\frac{a}{2}, 0\right)$ & radius $= \frac{a}{2}$.

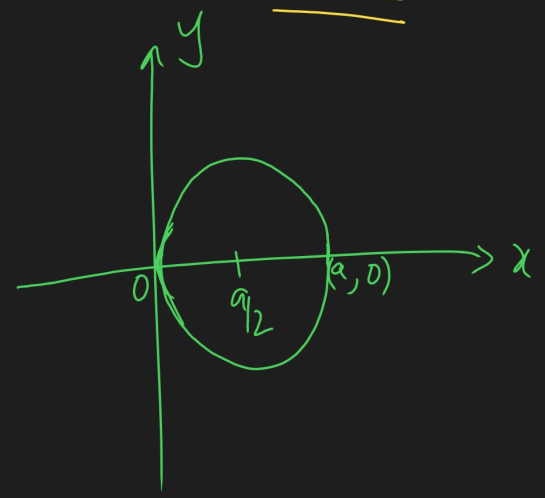
$ax^2+by^2+2hxy+2gx+2fy=0$

gen eqn of a circle

$(x-a)^2+(y-b)^2 = r^2$

is a circle with center (a,b) & radius r

$\left(x-\frac{a}{2}\right)^2 = x^2 - \cancel{2}\cdot x\cdot\frac{a}{\cancel{2}} + \frac{a^2}{4}$



2. Identify the curve

$$r = a\sin\theta$$

$$\Rightarrow r^2 = a^2\sin^2\theta$$

ie $x^2+y^2 = a(\underbrace{a\sin\theta}_{\downarrow})\sin\theta$

$$\Rightarrow x^2+y^2 = a\,\underline{r\sin\theta}$$

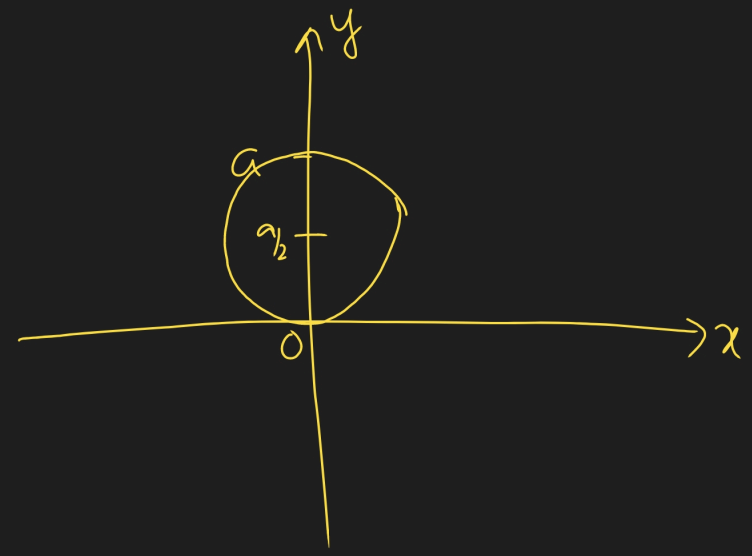
$$\Rightarrow x^2+y^2 = ay$$

$$\Rightarrow x^2+y^2-ay = 0$$

$$\Rightarrow x^2+\left(y-\frac{a}{2}\right)^2-\frac{a^2}{4} = 0$$

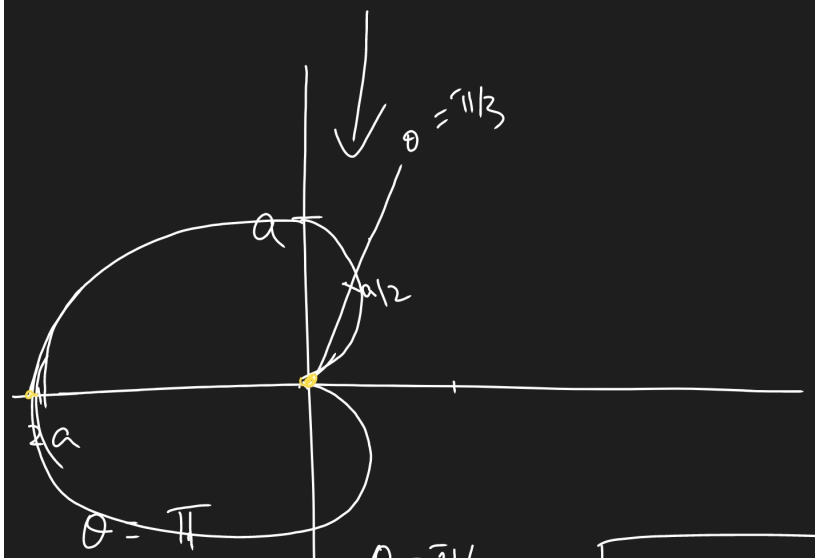
$$\Rightarrow x^2+\left(y-\frac{a}{2}\right)^2 = \left(\frac{a}{2}\right)^2$$

Which represents a circle with center $\left(0, \frac{a}{2}\right)$ & radius $\frac{a}{2}$.



Identify

3. $r = a(1-\cos\theta)$



$\theta = \pi$

$r = a(1-\cos\pi)$

$= a(1-(-1))$

$\boxed{r = 2a}$

$\theta = \pi/3$

$r = a(1-\frac{1}{2})$

$r = a/2$

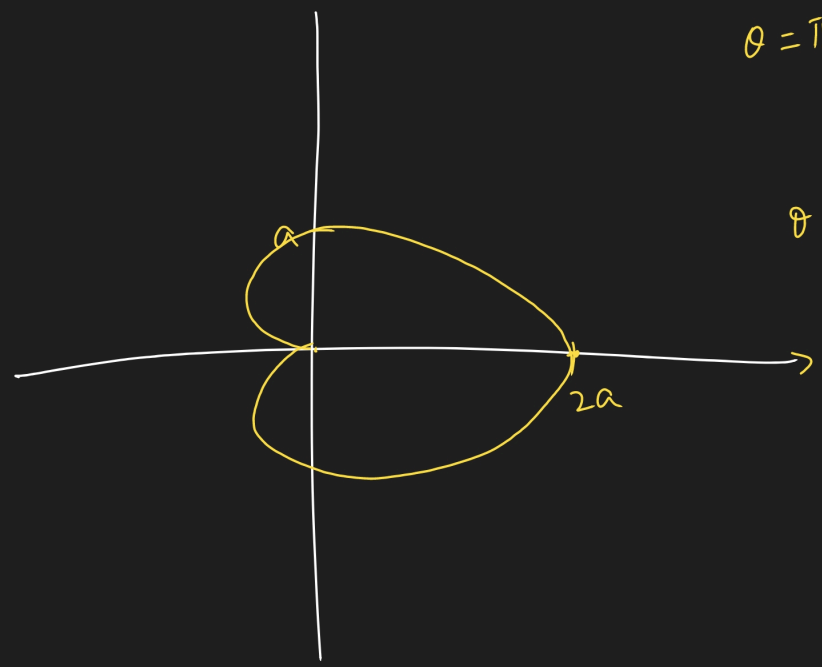
$\theta = 0,\ r = a(1-1)$

$= 0$

$\theta = \pi/2\ \ r = a(1-0)$

$= a$

4. $r = a(1+\cos\theta)$



$\theta = 0,\ r = a(1+\cos 0)$

$\Rightarrow r = 2a$

$\theta = \pi/2,\ r = a(1+\cos\pi/2)$

$\Rightarrow \boxed{r = a}$

$\theta = \pi,\ r = a(1+\cos\pi)$

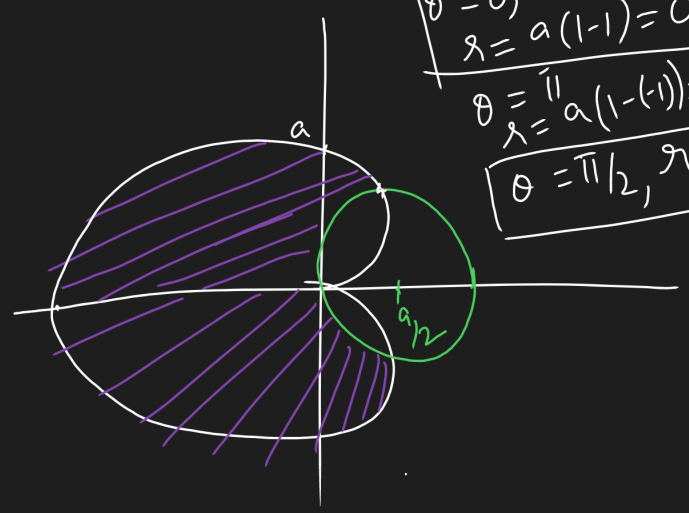
$= a(1-1)$

$r = 0$

Egs

① Find the length of the cardioid $r = a(1-\cos\theta)$ lying outside $r = a\cos\theta$.

Soln: We have,



$\theta = 0,\ r = a(1-1) = 0$

$\theta = \pi,\ r = a(1-(-1)) = 2a$

$\theta = \pi/2,\ r = a$

$$s = \int_{\theta=\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

To find θ at the point of intersection,

$$\cancel{a}(1-\cos\theta) = \cancel{a}\cos\theta$$

(The pt satisfies both the eqns)

$$\Rightarrow 1 - \cos\theta = \cos\theta$$

$$\Rightarrow 2\cos\theta = 1$$

$$\cos\theta = 1/2 \Rightarrow \boxed{\theta = \pm\pi/3}$$

Required

$$s = 2\int_{\theta=\pi/3}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$\underline{4a\sqrt{3}}$$

$$r = a(1-\cos\theta)$$

$$\frac{dr}{d\theta} = a(0-(-\sin\theta))$$

$$= a\sin\theta \quad //$$